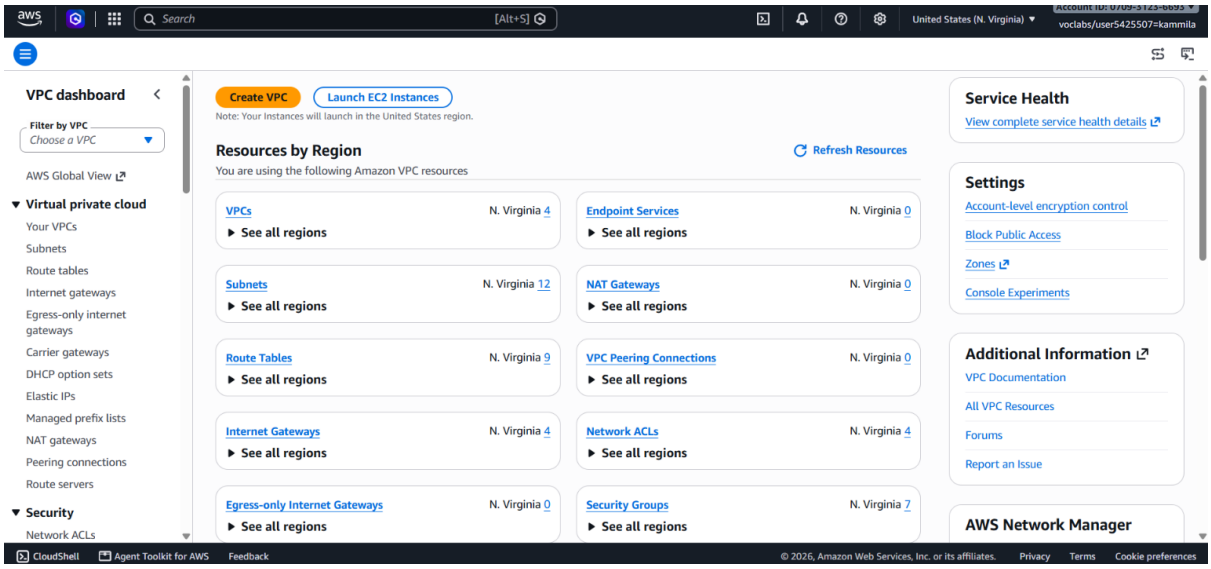
Practical 5

Build Your DB Server and Interact with Your DB Using an App.

Step 1: Create a Security Group for the RDS DB Instance

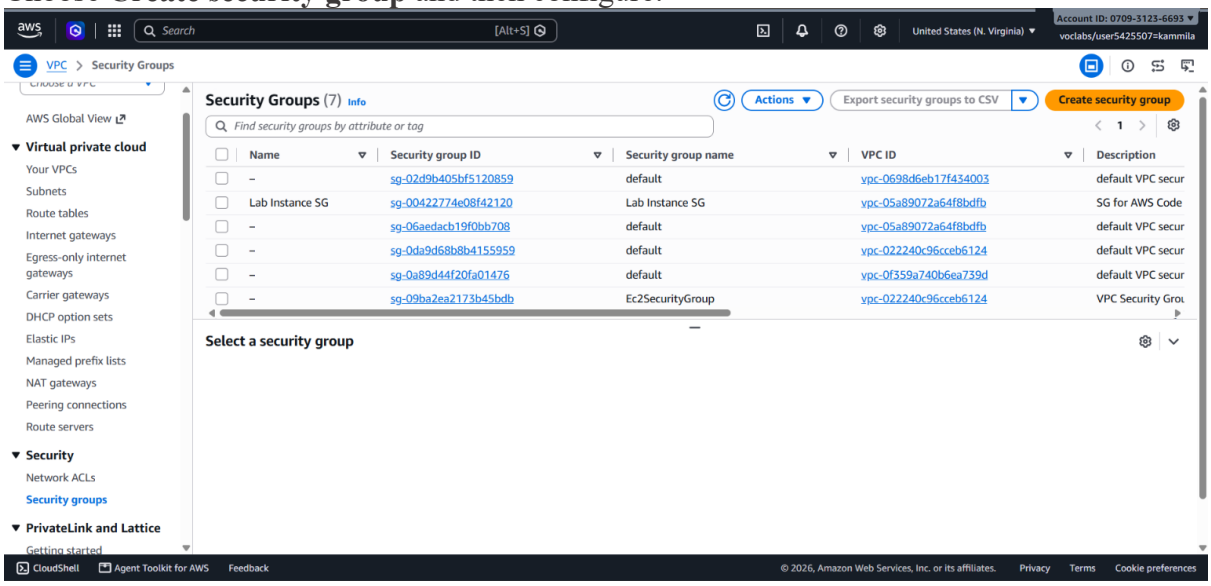
In this task, you will create a security group to allow your web server to access your RDS DB instance. The security group will be used when you launch the database instance.

- In the AWS Management Console, in the search box next to **Services,** search for and select **VPC**.

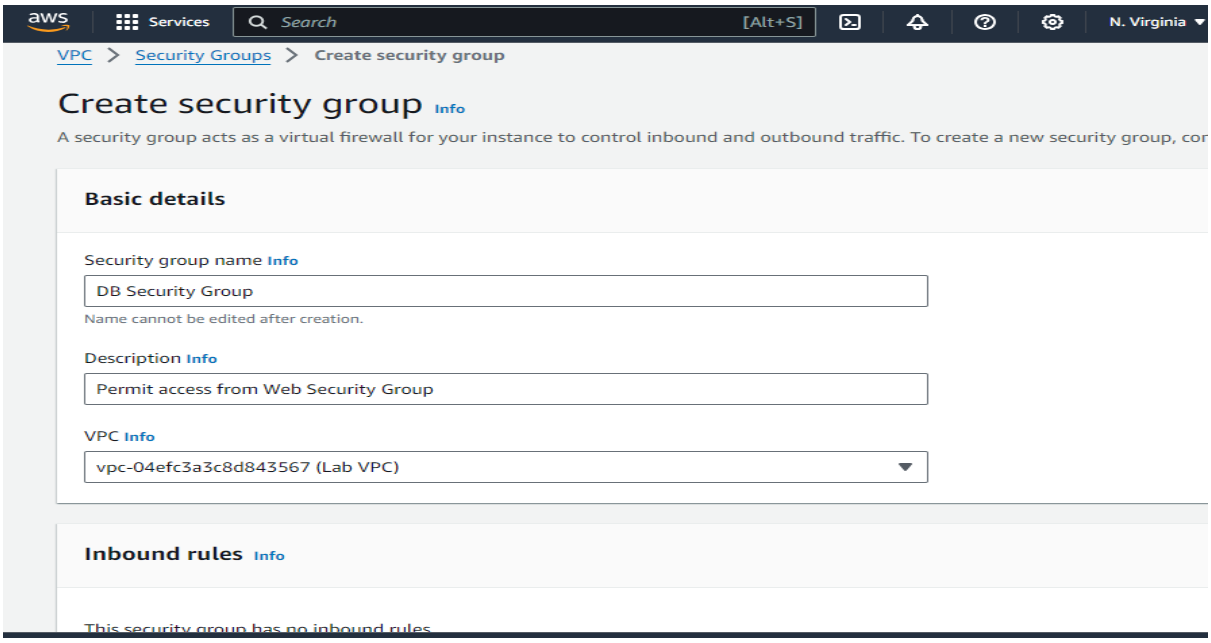


- In the left navigation pane, choose **Security groups**.

Choose **Create security group** and then configure:



Security group name: DB Security Group
Description: Permit access from Web Security Group
VPC: *Lab VPC*

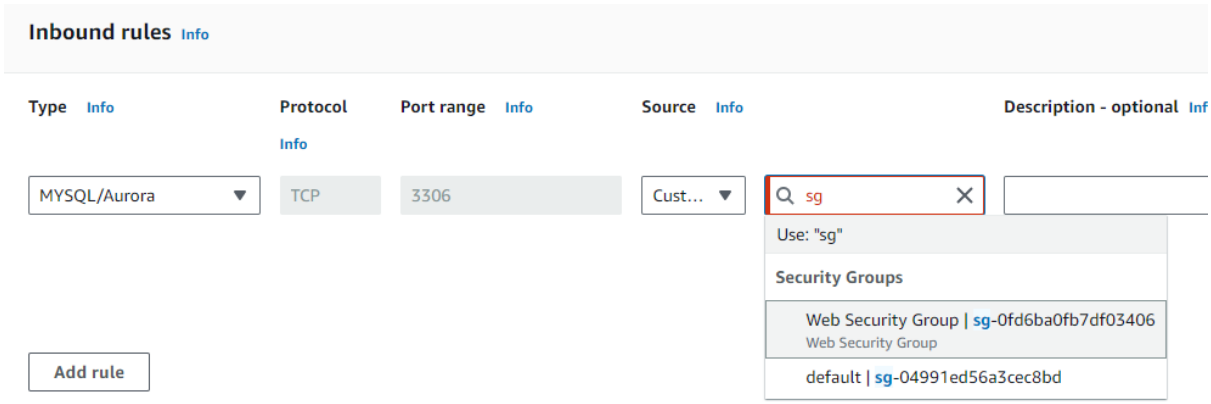


Tip: Choose the X next to VPC that is already selected, then choose **Lab VPC** from the menu.

- In the **Inbound rules** pane, choose **Add rule**

The security group currently has no rules. You will add a rule to permit access from the *Web Security Group*.
Configure the following settings:

- **Type:** *MySQL/Aurora (3306)*
- **Source:** Place your cursor in the field to the right of Custom, type sg, and then select *Web Security Group*.



This configures the Database security group to permit inbound traffic on port 3306 from any EC2 instance that is associated with the *Web Security Group*.

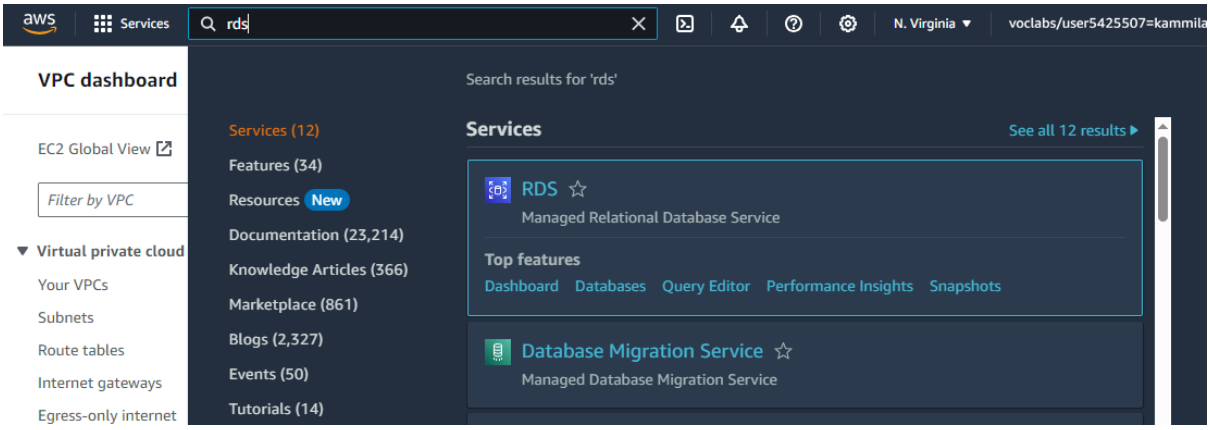
- Choose **Create security group**

You will use this security group when launching an Amazon RDS database in this lab.

Step 2: Create a DB Subnet Group

In this task, you will create a *DB subnet group* that is used to tell RDS which subnets can be used for the database. Each DB subnet group requires subnets in at least two Availability Zones.

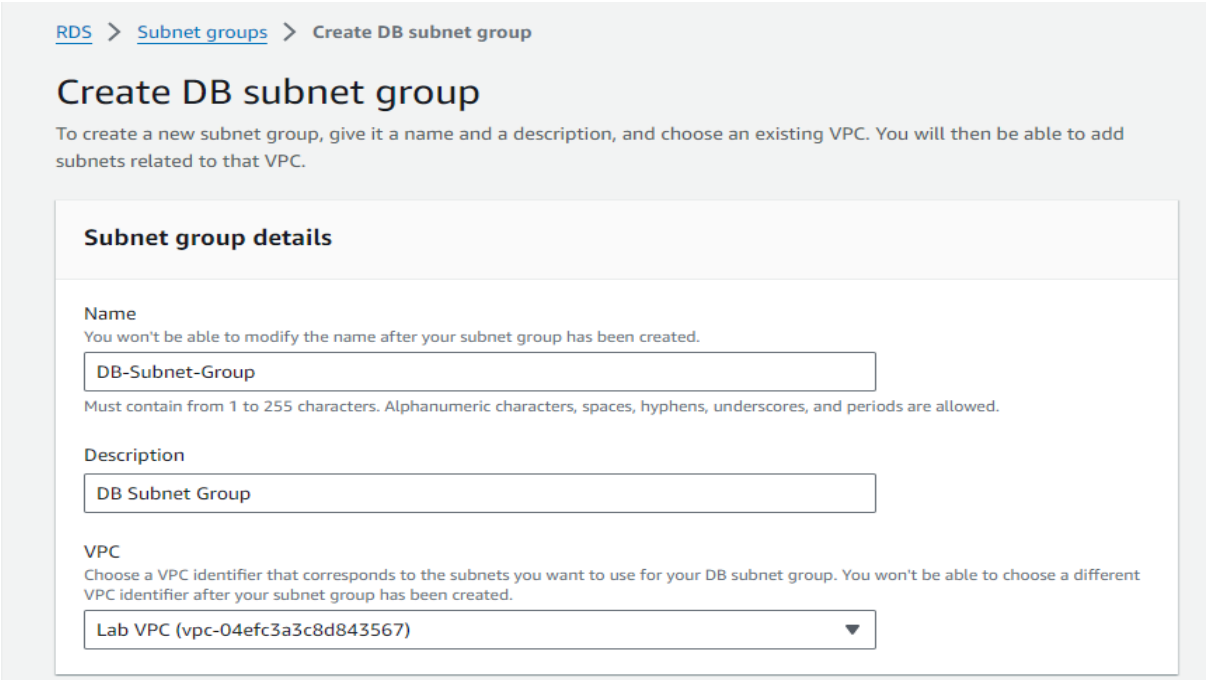
- In the AWS Management Console, in the search box next to **Services** , search for and select **RDS**.



- In the left navigation pane, choose **Subnet groups**.

If the navigation pane is not visible, choose the menu icon in the top-left corner.

- Choose **Create DB Subnet Group** then configure:
  - **Name:** DB-Subnet-Group
  - **Description:** DB Subnet Group
  - **VPC:** *Lab VPC*



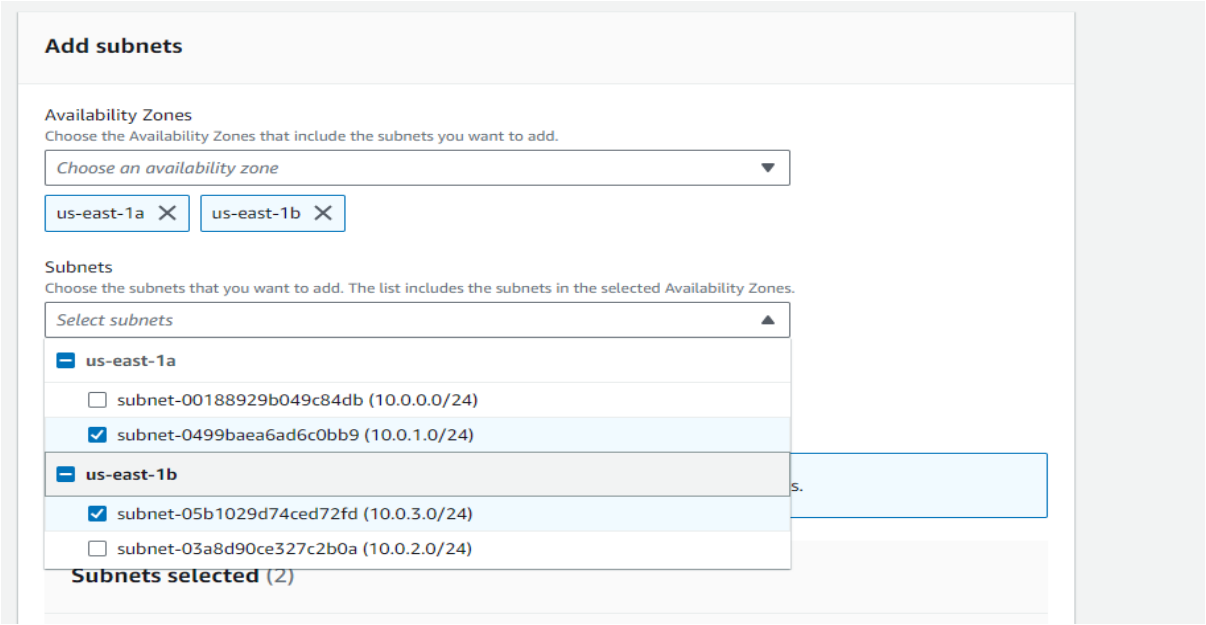
- Scroll down to the **Add subnets** section.

- Expand the list of values under **Availability Zones** and select the first two zones: **us-east-1a** and **us-east-1b**.
- Expand the list of values under **Subnets** and select the subnets associated with the CIDR ranges **10.0.1.0/24** and **10.0.3.0/24**.

These subnets should now be shown in the **Subnets selected** table.

- Choose **Create**

We will use this DB subnet group when creating the database in the next task.

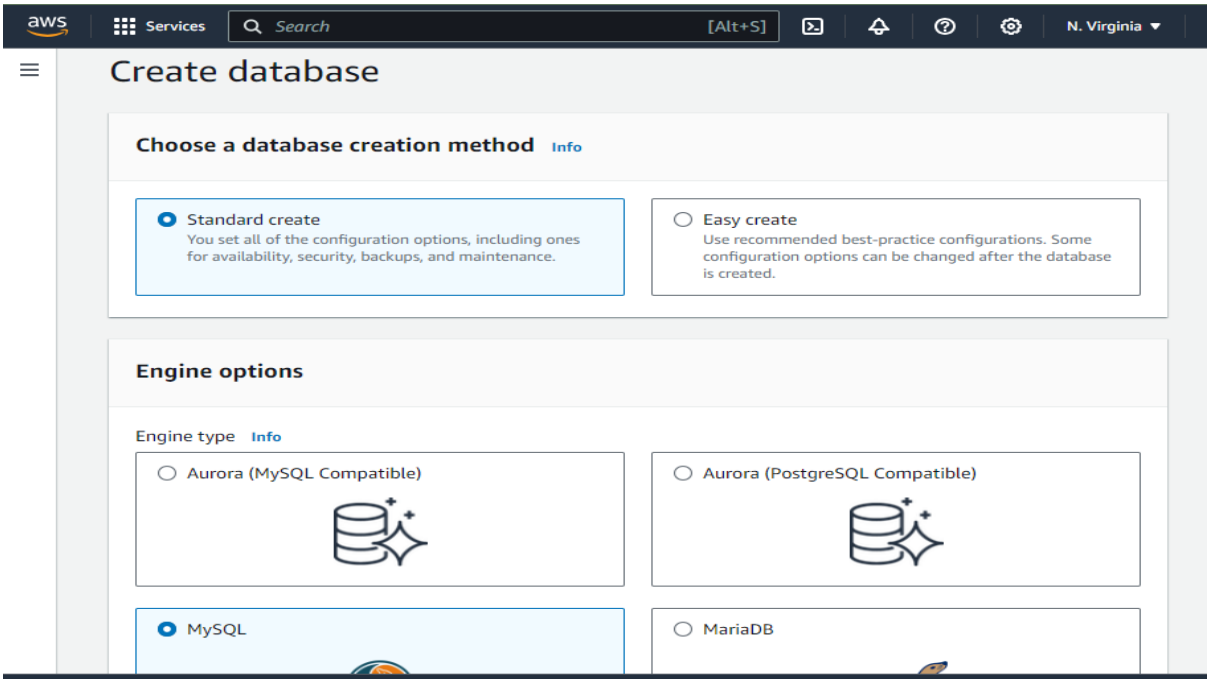


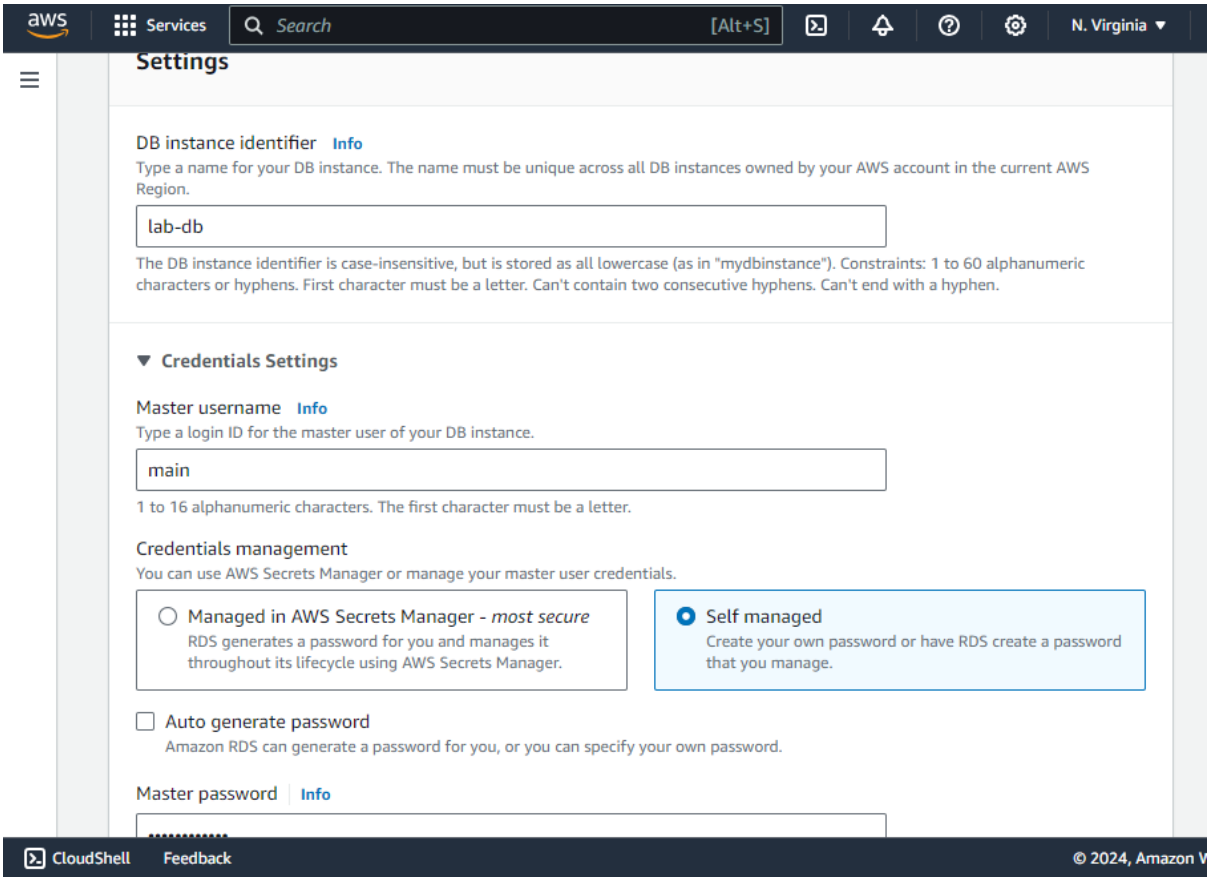
Step 3: Create an Amazon RDS DB Instance
In this task, we will configure and launch a Multi-AZ Amazon RDS deployment of a MySQL database instance.

- In the left navigation pane, choose **Databases**.
- Choose **Create database**

If we see **Switch to the new database creation flow** at the top of the screen, please choose it.

- Select **MySQL** under **Engine Options**.
- Under **Templates** choose **Dev/Test**.
- Under **Availability and durability** choose **Multi-AZ DB instance**.
- Under **Settings**, configure:
  - **DB instance identifier:** lab-db
  - **Master username:** main
  - **Master password:** lab-password
  - **Confirm password:** lab-password

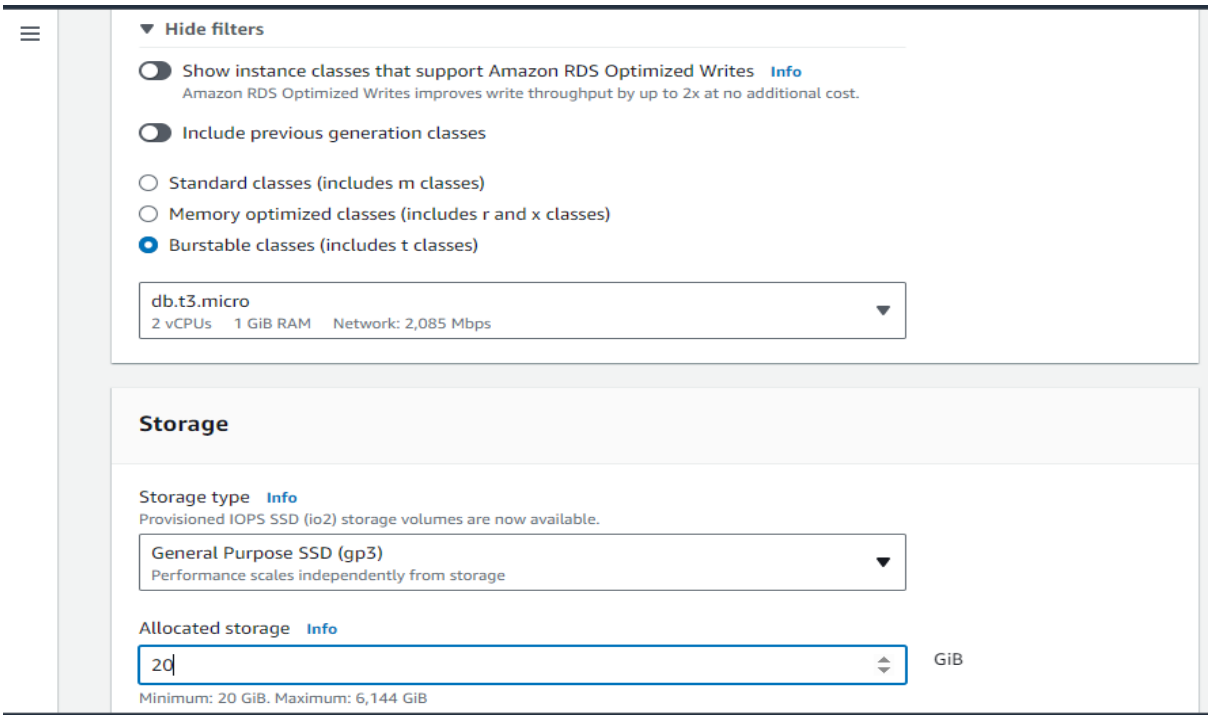




- Under **DB instance class**, configure:

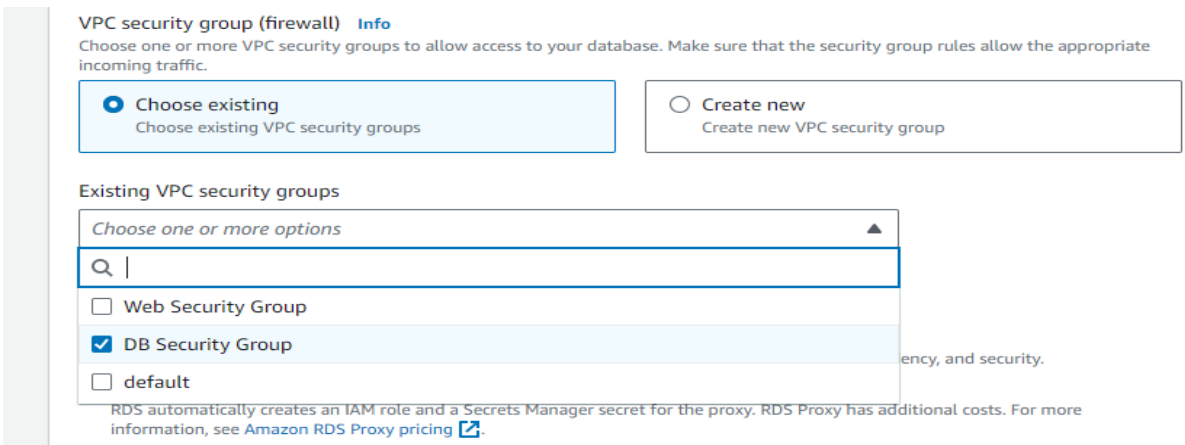
- Select **Burstable classes (includes t classes)**.
- Select *db.t3.micro*

- Under **Storage**, configure:
  - **Storage type:** *General Purpose (SSD)*
  - **Allocated storage:** *20*



Under **Connectivity**, configure:
- **Virtual Private Cloud (VPC):** *Lab VPC*

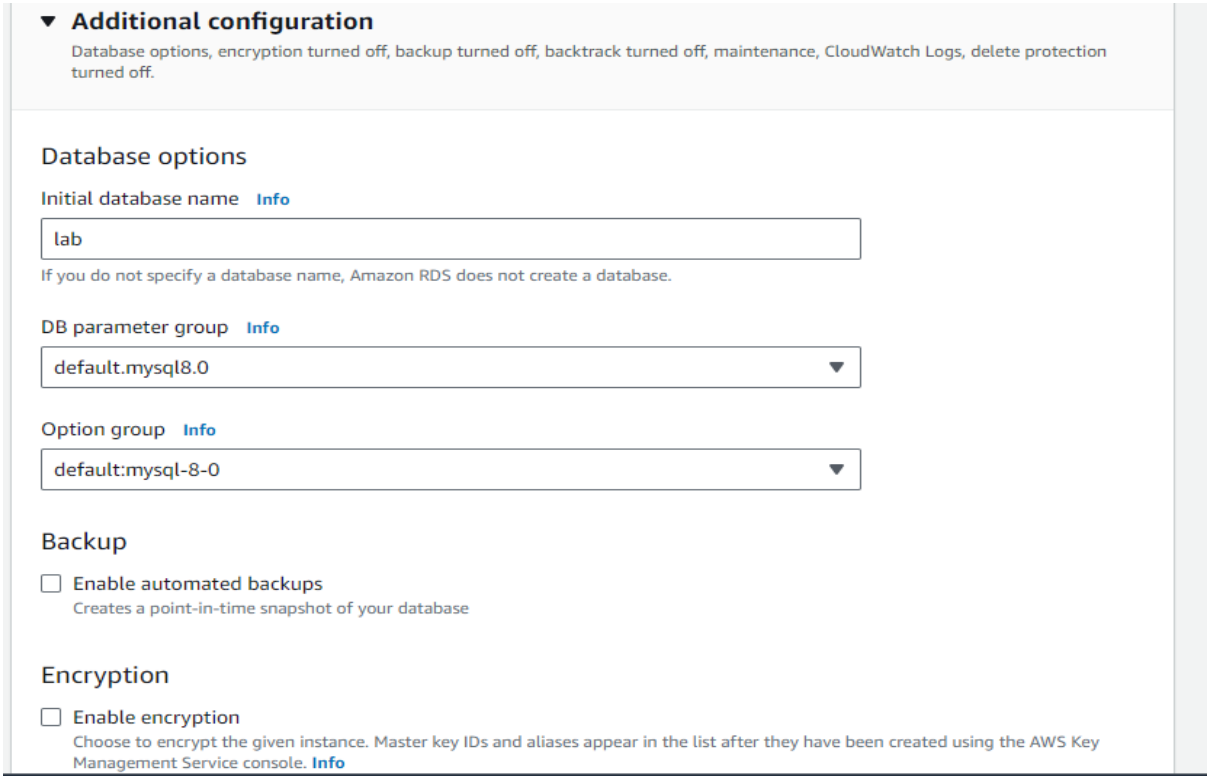
- Under **Existing VPC security groups**, from the dropdown list:
  - Choose *DB Security Group*.
  - Deselect *default*.



- Under **Monitoring** expand **Additional configuration**.

- Uncheck **Enable Enhanced monitoring**.

- Under **Additional configuration**, configure:
  - **Initial database name:** lab
  - Uncheck **Enable automatic backups**.
  - Uncheck **Enable encryption**



This will turn off backups, which is not normally recommended, but will make the database deploy faster for this lab.

- Choose **Create database**

Your database will now be launched.

- Wait until **Info** changes to **Modifying** or **Available**.
- Scroll down to the **Connectivity & security** section and copy the **Endpoint** field.

It will look similar to: *lab-db.xxxx.us-east-1.rds.amazonaws.com*.

- Paste the Endpoint value into a text editor. You will use it later in the lab.

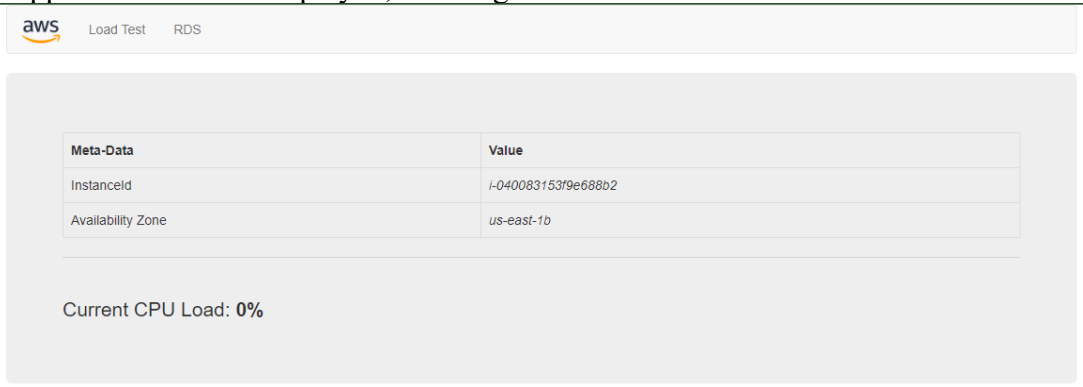
Step 4: Interact with Your Database

In this task, We will open a web application running on a web server that has been created for you. We will configure it to use the database that We just created.

- To discover the **WebServer** IP address, choose on the AWS Details drop down menu above these instructions. Copy the IP address value.

- Open a new web browser tab, paste the *WebServer* IP address and press Enter.

The web application will be displayed, showing information about the EC2 instance.

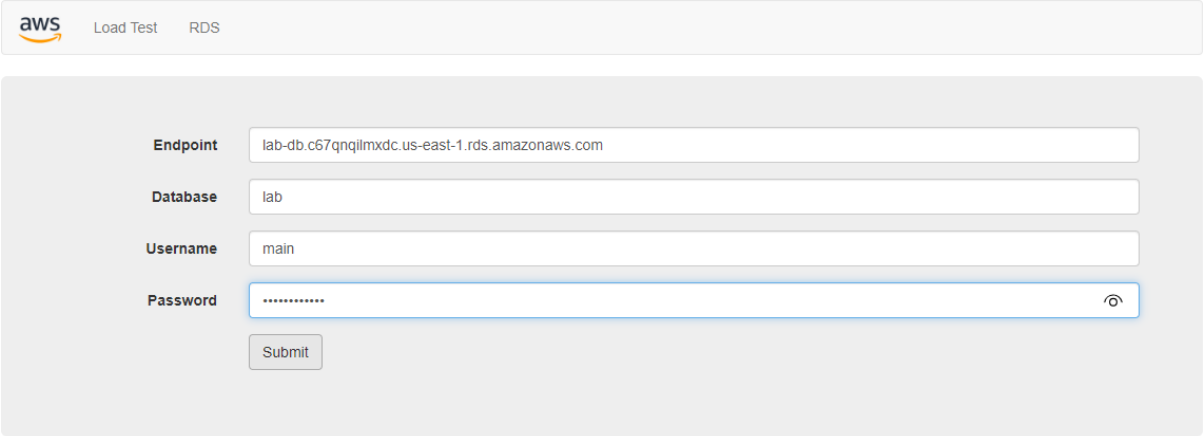


- Choose the **RDS** link at the top of the page.

We will now configure the application to connect to your database.

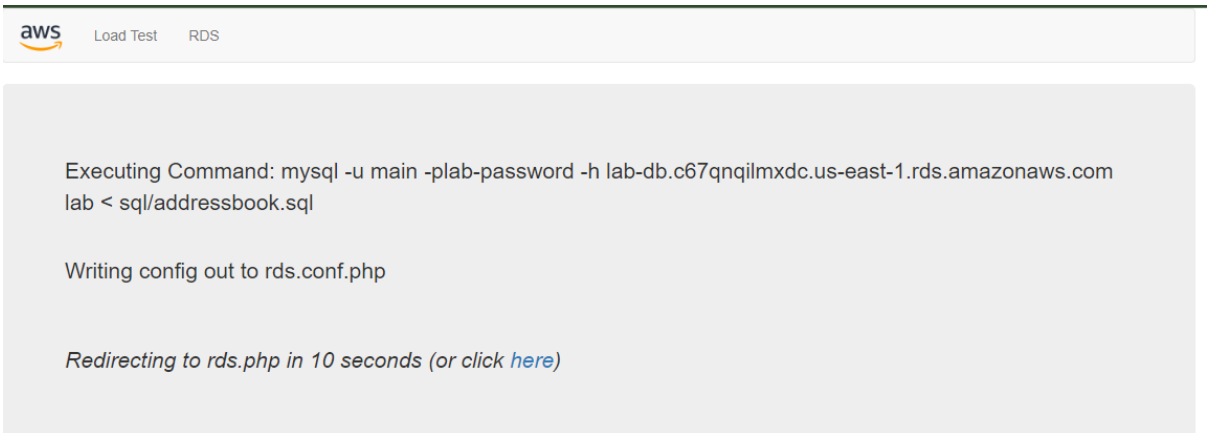


- Configure the following settings:
  - **Endpoint:** Paste the Endpoint We copied to a text editor earlier
  - **Database:** lab
  - **Username:** main
  - **Password:** lab-password
  - Choose **Submit**



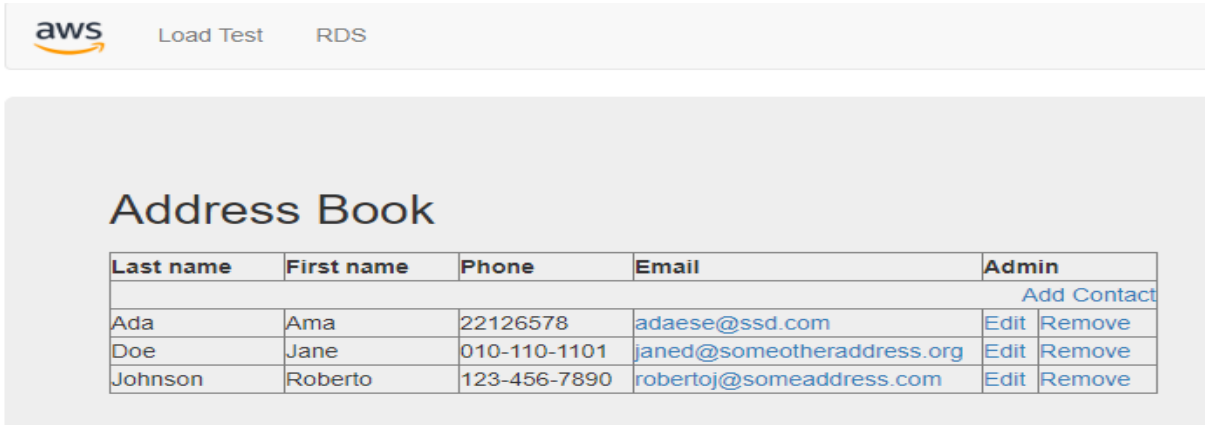
A message will appear explaining that the application is running a command to copy

information to the database. After a few seconds the application will display an **Address Book**. The Address Book application is using the RDS database to store information.



- Test the web application by adding, editing and removing contacts.





The data is being persisted to the database and is automatically replicating to the second Availability Zone.